

REVIEW

Role of Senescence in the Pathophysiology of Preeclampsia and Future Health

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ABSTRACT: Preeclampsia is a pregnancy-specific hypertensive disorder affecting up to 5% of pregnancies worldwide and is one of the leading causes of maternal and fetal morbidity and mortality globally. It is increasingly recognized that preeclampsia, from both clinical and pathophysiological standpoints, is a heterogeneous disease and that several mechanisms may lead to a clinical syndrome of hypertension, systemic disease, and proteinuria. Beyond its acute obstetric consequences, preeclampsia confers a significantly increased risk of future hypertension, cardiovascular disease, chronic kidney disease, and multimorbidity. Distinct underlying pathological mechanisms at the time of pregnancy may not only define clinical presentation and subtype of preeclampsia but may also make varying contributions to future cardiovascular and kidney disease. Emerging evidence implicates cellular senescence, a state of irreversible cell-cycle arrest accompanied by a proinflammatory senescence-associated secretory phenotype, as one of the mechanisms of preeclampsia that may serve as a mechanistic link between preeclampsia, accelerated cardiovascular aging, and future cardiovascular and kidney disease. This review synthesizes current evidence supporting the role of senescence in the pathophysiology of preeclampsia, demonstrated as accelerated epigenetic aging, increased senescence burden, and mesenchymal stem cell dysfunction, and discusses how persistence of senescence and propagation of senescent cells may contribute to vascular dysfunction decades after affected pregnancies. We further explore the translational implications of senolytic and senomorphic therapies as potential disease-modifying strategies in women with a history of preeclampsia.

Key Words: cardiovascular disease ■ heart failure ■ multimorbidity ■ proteinuria ■ risk factors

Preeclampsia: From Pregnancy Complication to Lifelong Cardiovascular Risk

Preeclampsia is a pregnancy-specific hypertensive disorder characterized by new-onset hypertension after mid-gestation, accompanied by proteinuria or evidence of maternal end-organ dysfunction.¹ Affecting ≈5% of pregnancies worldwide, preeclampsia remains a leading cause of maternal and perinatal morbidity and mortality globally. Despite increased awareness of preeclampsia heterogeneity and the complexity of its pathophysiology, the numerous mechanisms contributing to preeclampsia have not been addressed in a systematic manner.² This is likely one of the reasons, if not the main reason, why current research efforts focusing on single animal models, screening tests, or therapies have failed to demonstrate universal diagnostic and treatment efficacy and efficiency across the spectrum of preeclampsia clinical presentations. Historically regarded as a disorder confined

to pregnancy, preeclampsia is now firmly established as an independent, sex-specific risk factor for future cardiovascular disease (CVD), chronic hypertension, stroke, heart failure, and chronic kidney disease³ (Figure 1). This excess risk manifests early, often within a decade of the affected pregnancy, and persists across the lifespan.

The recognition of preeclampsia as a cardiovascular risk state has fundamentally altered how pregnancy is viewed within cardiovascular medicine. Pregnancy is increasingly conceptualized as a physiological stress test, capable of unmasking latent cardiovascular vulnerability, which could result in hypertensive disorders of pregnancy and CVD at different times in a woman's life. Alternatively, preeclampsia itself may induce durable vascular injury, thus increasing the risk of future CVD and chronic kidney disease. Despite strong epidemiological evidence linking preeclampsia to future disease, the biological mechanisms, similar to the disease itself, remain incompletely defined. Research to date has focused on

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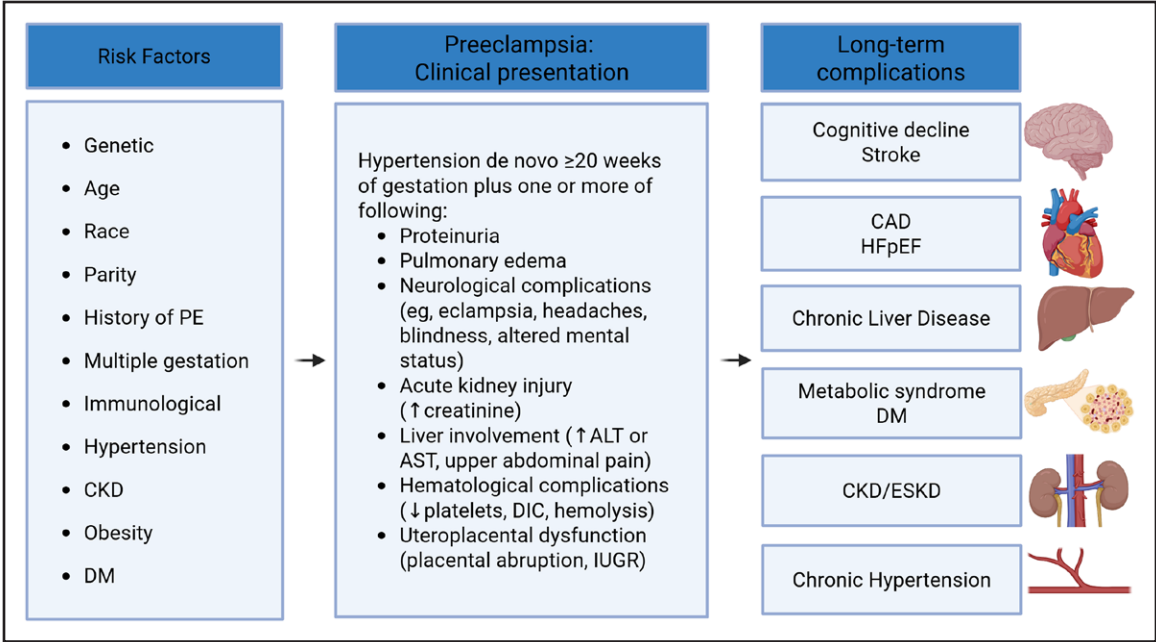


Figure 1. Preeclampsia (PE) and related long-term complications: the role of shared risk factors. ALT indicates alanine transaminase; AST, aspartate aminotransferase; CAD, coronary artery disease; CKD, chronic kidney disease; DIC, disseminated intravascular coagulation; DM, diabetes; ESKD, end-stage kidney disease; HFpEF, heart failure with preserved ejection fraction; and IUGR, intrauterine growth restriction.

shared cardiometabolic risk factors, residual endothelial dysfunction, or incomplete postpartum recovery.

An emerging body of evidence suggests that preeclampsia may initiate or accelerate fundamental biological aging processes. Among these, cellular senescence, a stress response characterized by irreversible cell-cycle arrest and a proinflammatory senescence-associated secretory phenotype (SASP; Figure 2), has gained attention as a unifying mechanism linking pregnancy complications to accelerated vascular aging. Cellular senescence is increasingly recognized as a driver of chronic disease, multimorbidity, and functional decline.⁴

The central premise of this review is that preeclampsia is mechanistically related to a senescent phenotype across multiple maternal tissues, and that this senescence persists and propagates beyond pregnancy, contributing to later-life CVD. We present current evidence that supports the role of senescence in the pathophysiology of preeclampsia and related long-term adverse cardiac and renal effects. It is based on the 2024 Arthur C. Corcoran Memorial Lecture that integrated historical and contemporary perspectives to examine how preeclampsia fits into a life-course model of cardiovascular and kidney disease in the affected women.

Cellular Senescence: Molecular Pathways Relevant to Hypertension and Vascular Aging

Cellular senescence is a physiological, irreversible cell-cycle arrest mechanism that plays several important roles in biological functions, including embryonic development,

placental formation, and the onset of labor in human pregnancy.⁵ Thus, senescence may be advantageous by promoting tissue repair and regeneration. However, evoked in response to stress, the irreversible proliferation arrest and failure to immunologically clear these cells lead to systemic metabolic and functional decline, and a premature aging-like phenotype over time. Senescence is a complex, multistep biological process triggered by diverse forms of cellular stress.⁶ Canonical inducers include DNA damage, telomere attrition, oxidative and metabolic stress, mitochondrial dysfunction, inflammatory signaling, oncogene activation, and mechanical forces such as shear stress. These stressors converge on highly conserved molecular pathways that induce cell-cycle arrest, primarily through activation of the p53–p21 and p16^{INK4a} axes.

Proliferative arrest, due to DNA damage and in response to noxious stimuli, such as hemodynamic shear stress, oxidative stress, inflammatory cytokines, and metabolic alterations, all of which have known contributions to the pathophysiology of preeclampsia, is a central initiating event in senescence. Persistent DNA double-strand breaks activate ATM and ATR kinases, leading to stabilization of p53 and induction of p21, a cyclin-dependent kinase inhibitor that enforces G1/S arrest. In parallel, or sequentially, stress-induced upregulation of p16^{INK4a} occurs. Although p21-mediated arrest may be reversible under some conditions, p16^{INK4a}-driven senescence typically is irreversible and associated with long-term persistence of senescent cells.

Beyond cell-cycle arrest, senescent cells undergo profound metabolic, epigenetic, and secretory

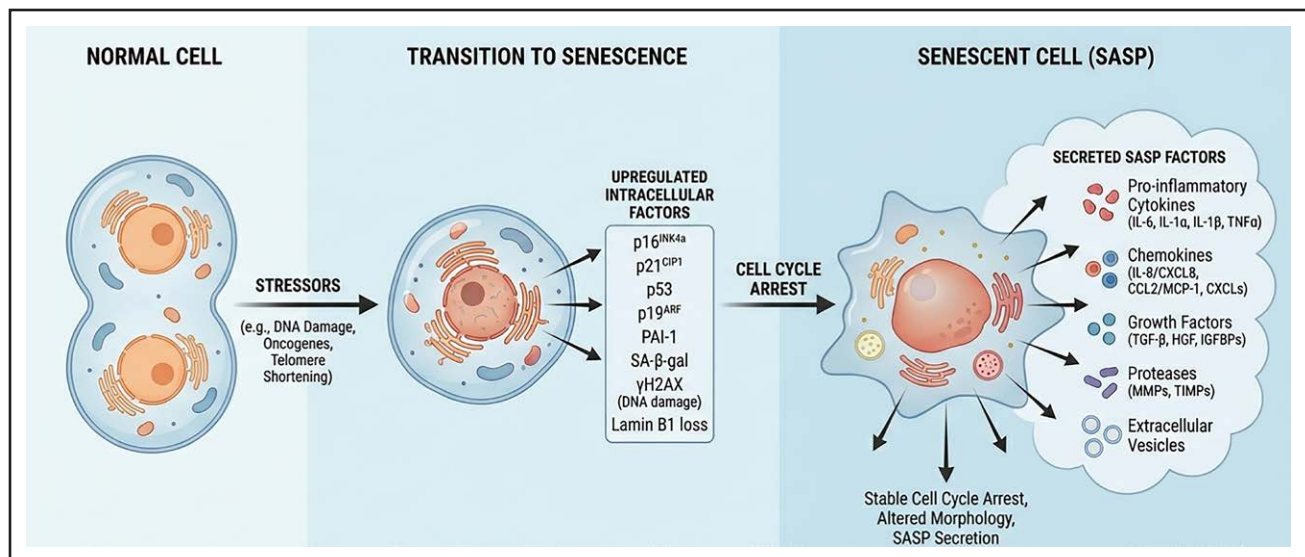


Figure 2. Cellular senescence: irreversible cell-cycle arrest mechanistically related to aging.

Normal cell, capable of undergoing cell-division (**left**) transitions to a senescent cell (**middle**) in response to DNA damage due to several noxious stimuli, which ultimately results in cell cycle arrest (**right**), altered morphology, and senescence-associated secretory phenotype (SASP) production. CCL2 indicates chemokine (C-C motif) ligand 2; CXCL, C-X-C motif chemokine ligands; CXCL8, C-X-C motif chemokine ligand 8, also known as IL-8; HGF, hepatocyte growth factor; IGFBP, insulin-like growth factor-binding protein; IL, interleukin; MCP-1, monocyte chemoattractant protein 1; MMP, matrix metalloproteinases; SA-β-gal, senescence-associated β-galactosidase; TGF-β, transforming growth factor β; TIMP, tissue inhibitors of metalloproteinase; TNFα, tumor necrosis factor alpha; and γH2AX, gamma phosphorylated histone H2AX. Figure 2 was created using Gemini; the corresponding author confirms that everything is correct in the figure.

reprogramming. Mitochondrial dysfunction plays a key role, with altered mitochondrial dynamics, increased reactive oxygen species production, and impaired mitophagy creating a self-amplifying loop of oxidative stress. Senescent cells also exhibit extensive chromatin remodeling, including formation of senescence-associated heterochromatin foci and global changes in DNA methylation patterns that resemble accelerated biological aging.

A defining feature of senescence is SASP.^{6,7} The SASP consists of cytokines, chemokines, growth factors, proteases, and extracellular matrix components. Core SASP factors include interleukin (IL)-6, interleukin-8, tumor necrosis factor-α, monocyte chemoattractant protein-1, plasminogen activator inhibitor-1, and matrix metalloproteinases. SASP factors exert pleiotropic effects: they reinforce senescence in an autocrine manner, induce senescence in neighboring cells (bystander senescence), promote chronic inflammation, impair angiogenesis, and disrupt tissue architecture.

Importantly, senescent cells are not inert. They actively resist apoptosis through upregulation of senescent cell antiapoptotic pathways, including BCL-2 family proteins, PI3K/AKT signaling, and tyrosine kinase pathways. This resistance allows senescent cells to accumulate with age and persist after acute insults, contributing to chronic tissue dysfunction.

Within the cardiovascular system, senescence has been implicated in endothelial dysfunction,⁸ vascular stiffening, impaired nitric oxide bioavailability, inflammation-driven atherogenesis, and reduced reparative capacity.

Senescent endothelial cells exhibit decreased proliferation, altered barrier function, and heightened inflammatory signaling, while senescent vascular smooth muscle cells contribute to arterial remodeling and calcification. These processes are central to the pathogenesis of hypertension and CVD, positioning senescence as a plausible mechanistic link between pregnancy-related stressors and later-life vascular pathology.⁹

Pregnancy as a Prosenescent Stress Test

Pregnancy represents a unique physiological state characterized by profound hemodynamic, metabolic, and immunologic adaptations. Cardiac output increases by up to 50%, systemic vascular resistance decreases, and blood volume expands substantially. These changes impose significant shear stress on the maternal vasculature, requiring robust endothelial plasticity and angiogenic capacity. Simultaneously, pregnancy necessitates finely tuned immune tolerance and metabolic flexibility to support fetal growth.¹⁰

Preeclampsia disrupts these adaptations. The disorder is characterized by abnormal placentation, impaired spiral artery remodeling, and placental ischemia, leading to increased oxidative stress and release of antiangiogenic and inflammatory factors into the maternal circulation. These circulating factors induce widespread endothelial dysfunction, inflammation, and vascular injury, which, in turn, are key triggers of senescence.

From a senescence perspective, pregnancy, in general, and preeclampsia, in particular, may represent a

perfect storm in which multiple senescence-inducing stressors converge (Figure 3). Notably, several physiological processes that are exaggerated in pregnancy, such as procoagulatory and proinflammatory states and oxidative stress, are further exacerbated in preeclampsia. Therein, hemodynamic shear stress, oxidative stress, inflammatory cytokines, metabolic alterations, and angiogenic imbalance act in concert to activate DNA damage responses and senescence pathways. On the contrary, the immunologic adaptations of pregnancy, including relative immune tolerance, may impair clearance of senescent cells. Preeclampsia-associated senescence may occur in stem and progenitor cell populations, such as mesenchymal stem cells (MSCs; see below), potentially compromising long-term tissue repair capacity. These features distinguish pregnancy/preeclampsia-induced senescence from senescence induced by transient injuries and may explain its persistence beyond delivery.

Evidence for Senescence and Accelerated Biological Aging in Preeclampsia

Several lines of evidence support the role of senescence in the pathophysiology of preeclampsia. First, preeclampsia placentas exhibit signs of premature aging, as evidenced

by ischemic changes and infarcts long before full-term is reached. Second, aberrant placental senescence has been shown in preeclampsia, with studies showing telomere shortening and increased expression of markers of senescence (p21 and p53) in preeclampsia placentas.^{11,12} Third, several SASP components, including IL-6 and IL-8, are elevated in women with preeclampsia versus normotensive pregnancies, both at the time of delivery and years after the affected pregnancies.^{13–15} Fourth, metformin, a senomorphic agent that attenuates SASP,⁶ has been shown to decrease the incidence of preeclampsia.¹⁶ This effect was considered to be due to its beneficial vascular and metabolic effects, but its senomorphic properties were not considered. Fifth, a recent study indicated that placental senescence in preeclampsia and circulating SASP may contribute to cardiac dysfunction in pregnancy and the development of postpartum cardiomyopathy.¹⁷ Sixth, increased syncytiotrophoblast senescence appears weeks before the clinical manifestations of early onset preeclampsia.¹⁸ This premise is supported by published studies from our laboratory, as discussed below.

Epigenetic Aging

DNA methylation-based epigenetic clocks provide a quantitative measure of biological aging that predicts

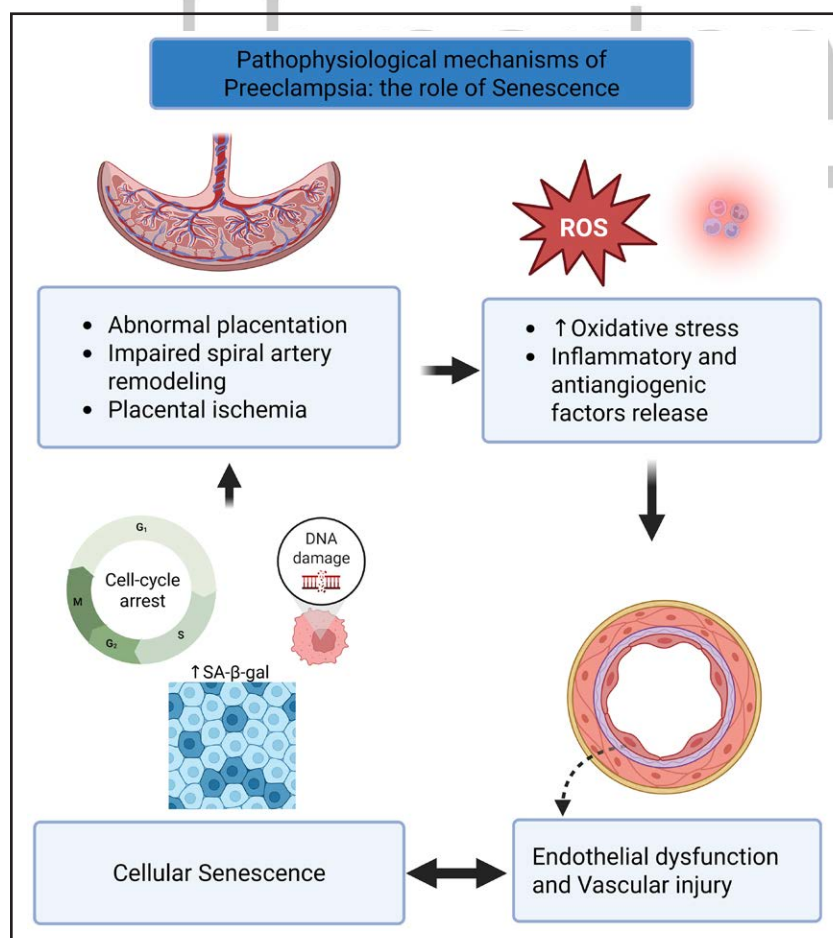


Figure 3. The role of senescence in the pathophysiology of preeclampsia. SA-β-gal indicates senescence-associated beta-galactosidase.

morbidity and mortality independent of chronological age.¹⁹ Our study of epigenetic clocks in pregnant populations has revealed striking differences between normotensive and preeclamptic pregnancies: women with preeclampsia exhibit progressive acceleration of epigenetic age across gestation, whereas epigenetic age remains stable throughout pregnancy in normotensive women.²⁰

At delivery, women with preeclampsia demonstrate a significant divergence between epigenetic and chronological age, consistent with pregnancy-associated acceleration of biological aging. This finding is notable because epigenetic age acceleration has been linked to CVD, hypertension, and all-cause mortality in non-pregnant populations. These data suggest that preeclampsia does not merely reveal preexisting aging phenotypes but actively accelerates aging trajectories during pregnancy.

Mechanistically, epigenetic aging during preeclampsia may reflect cumulative exposure to oxidative stress, inflammation, and metabolic dysregulation. Importantly, epigenetic modifications are durable and may persist long after pregnancy, providing a molecular substrate for long-term disease risk.

Circulating and Urinary Markers of Senescence

Beyond epigenetic clocks, we have demonstrated that women with preeclampsia demonstrate increased systemic senescence burden.²⁰ Circulating levels of p16^{INK4a}, a canonical marker of senescence, are elevated in preeclamptic pregnancies compared with normotensive controls. While p21 levels may not differ consistently, increased p16 expression suggests activation of irreversible senescence pathways.

Urinary α -Klotho levels are reduced in women with preeclampsia. α -Klotho is a multifunctional protein with antiaging, antioxidative, and vasculoprotective properties.²¹ Reduced α -Klotho has been associated with endothelial dysfunction, hypertension, and chronic kidney disease. Its reduction in preeclampsia supports the presence of a systemic proaging environment.

Tissue-Level Evidence of Senescence

We performed direct tissue analyses that provided compelling evidence for senescence in preeclampsia. Increased expression of p16^{INK4a} was observed in renal and adipose tissue from women with preeclampsia compared with normotensive pregnancies.²⁰ The kidney is a critical target organ in preeclampsia and a key determinant of long-term cardiovascular and renal risk. Senescence within renal tissue may contribute to persistent glomerular and vascular dysfunction, predisposing affected women to chronic kidney disease and hypertension.

Adipose tissue, increasingly recognized as an active endocrine and inflammatory organ, also exhibited increased senescence in preeclampsia. Adipose tissue

from preeclamptic pregnancies demonstrates upregulation of SASP components, including monocyte chemoattractant protein-1 and tumor necrosis factor- α . Senescent adipose tissue may serve as a reservoir of chronic inflammation, amplifying systemic vascular risk.

Critical evidence has come from studies of preeclampsia placentas, as the placenta plays a central role in preeclampsia pathophysiology. A recent study of preeclampsia placentas demonstrated increased levels of multiple markers of senescence and tissue aging, as well as overall increased gene expression of 28 circulating SASP proteins in serum samples from patients with preeclampsia.¹⁷ Another study about the placental cell type involved examined the human maternal-fetal interface (fetal villi and maternal decidua) using single-nucleus RNA sequencing combined with spatial transcriptomics.¹⁸ Results of the study showed that dysregulation of the syncytiotrophoblast due to premature senescence appears weeks before the clinical manifestations of early onset preeclampsia (27–33 weeks of gestation).

Collectively, these findings indicate that preeclampsia is associated with widespread senescence affecting multiple maternal tissues, and the placenta. In 2022, the Cellular Senescence Network, a National Institutes of Health Common Fund initiative, was established to map senescent cells across the human lifespan and different organs, including the placenta, to advance diagnostic and therapeutic approaches to improve human health.²² Placental samples from women 34 years of age or younger, and those 35 years or older, will be analyzed to map senescent cells using leading-edge technologies and advanced computational methodologies, including single-nucleus RNA-seq, single-cell RNA-seq, imaging mass cytometry, and spatial transcriptomics. This research will lay the groundwork for future studies aimed at treating reproduction-related pathologies, including preeclampsia, through targeting/modulating the process of senescence.

Mesenchymal Stem Cells, Angiogenesis, and Vascular Repair Failure

Mesenchymal stem cells play a central role in vascular homeostasis, angiogenesis, immune modulation, and tissue repair. During normal pregnancy, MSCs support placental development and maternal vascular adaptation through paracrine signaling and direct cellular interactions.

We studied MSCs that were isolated from abdominal fat tissue explants removed during medically indicated C-sections from preeclampsia women and compared them to MSCs obtained from normotensive pregnancies. We demonstrated that MSC function is impaired in preeclampsia: MSCs isolated from women with preeclampsia showed reduced angiogenic potential, particularly under proinflammatory conditions.²³ In addition,

preeclampsia-MSCs demonstrated reduced proliferative capacity and a senescence phenotype, as reflected by more abundant staining for senescence-associated beta galactosidase and upregulation of senescence markers and SASP components.

MSC senescence may have long-term consequences beyond pregnancy. MSCs contribute to vascular repair and regeneration throughout life; thus, senescence-induced impairment of MSC function during pregnancy may compromise postpartum vascular recovery and predispose the affected women to future CVD.

Preclinical studies

We have adopted and established a preeclampsia animal model using IL-10 knockout (IL-10^{-/-}) mice. This strain, in a nonpregnant state, demonstrates an increased burden of senescent cells compared with wild-type controls.²⁴ When injected intraperitoneally with human preeclampsia sera on gestational day 10,²⁵ the IL-10^{-/-} model fully replicates the clinical features of preeclampsia, including hypertension, proteinuria, abnormal vascular reactivity, foot process effacement and endotheliosis in kidney sections, and a known imbalance between antiangiogenic sFlt-1 (from fms-like tyrosine kinase receptor-1) and proangiogenic PlGF (placental growth factor), favoring the former. The model is fully dependent on pregnancy status, as hypertension and proteinuria develop only in pregnant, but not in nonpregnant IL-10^{-/-} mice, thus further mimicking human disease. This is in sharp contrast with other animal models, whereby most interventions result in hypertension and proteinuria not only in nonpregnant mice, but in male mice as well.²⁶ Similarly, the model is dependent on an IL-10^{-/-} state: wild-type mice receiving preeclampsia sera on gestational day 10 do not develop preeclampsia-like features.

Critical features of the IL-10^{-/-} preeclampsia model deserve additional consideration, particularly with respect to clinical correlations to human disease. The presence of an IL-10^{-/-} state mirrors the human disease, with our systematic review and meta-analysis showing that women with preeclampsia have decreased IL-10 levels compared with normotensive controls.²⁷ Decreased IL-10 levels are likely, at least in part, due to the known reduction in numbers and impaired function of Treg (regulatory T) cells in preeclampsia.²⁸ The deficiency in IL-10-mediated immunosuppression may contribute to heightened inflammatory responses, endothelial dysfunction, and abnormal placental development in preeclampsia. On the contrary, it is somewhat counterintuitive to postulate that senescence, a mechanism of aging, plays a role in the pathophysiology of preeclampsia in a subset of young women (<18 years). Young age, even in the absence of any significant comorbidities, consistently has been identified as a major risk factor for preeclampsia. Under these circumstances, a preeclampsia pregnancy may, indeed, trigger senescence. In contrast, in

women with known preeclampsia risk factors, associated with increased senescent burden (eg, diabetes, obesity, kidney disease), a prosenescence state may predate pregnancy. Future research, both preclinical and clinical, should address these critical questions, specifically in the context of the heterogeneity of preeclampsia, whereby the spectrum of clinical presentations implies distinct underlying pathological mechanisms.

Senescence Burden, Propagation and Multimorbidity

One mechanism that might underpin the relationship between preeclampsia and future CVD is that these conditions share several common risk factors (eg, obesity, diabetes, and renal disease). It is particularly intriguing to postulate that preeclampsia may induce long-term metabolic and vascular abnormalities that might increase the overall risk of CVD later in life. In the case of a senescence phenotype, this may be due to its ability to persist in the affected tissues and propagate to remote sites. This has been demonstrated by experimental intraperitoneal transplantation studies of irradiated, senescent preadipocytes.²⁴ The transplanted mice demonstrated reduced physical performance, frailty, and persistent tissue dysfunction long after transplanted cells were cleared and no longer detectable.²⁴ Indeed, SASP factors can induce senescence in neighboring cells, creating a vicious circle that amplifies tissue dysfunction. This results in sustained inflammation, functional decline, and multimorbidity. These findings challenge the notion that senescence is a localized or transient phenomenon and instead support a model in which senescence can spread both locally and systemically and exert long-lasting effects.

Clinical correlates of these experimental findings are epidemiological studies showing that a history of hypertensive disorders of pregnancy is associated with increased multimorbidity,^{29,30} including cardiovascular, metabolic, and renal diseases. This clustering of chronic conditions is consistent with accelerated biological aging³¹ and supports the concept that pregnancy-associated senescence may initiate a cascade of metabolic and vascular abnormalities, leading to widespread disease vulnerability.

Senolytics as Proof of Mechanistic Causality

Senolytic agents selectively eliminate senescent cells by targeting their anti-apoptotic survival pathways.^{6,32} Dasatinib, a pan-tyrosine kinase inhibitor with established clinical use, has emerged as a potent senolytic in preclinical studies. Short-term exposure to dasatinib induces apoptosis preferentially in senescent cells while sparing nonsenescent cells.

Dasatinib treatment of senescent MSCs obtained from preeclamptic women decreases the senescence

burden in and restores the angiogenic capacity of preeclampsia-MSCs to levels comparable to MSCs from normotensive pregnancies.²³ This restoration provides critical proof-of-concept that senescence directly mediates MSC dysfunction and that its elimination reverses key pathological features.

These findings support a causal role for senescence in preeclampsia-related vascular impairment rather than a purely associative relationship. They also suggest that at least some components of pregnancy-induced vascular damage may be reversible and that mechanism-specific therapies may be available.

Therapeutic and Preventive Implications

The recognition of senescence as a mechanistic link between preeclampsia and later-life CVD opens new translational opportunities, with either senolytic therapies that could reduce senescent cell burden or senomorphic agents that may attenuate SASP release and signaling without eliminating senescent cells.³³

As the process of senescence is critical for embryogenesis, senolytics are contraindicated during pregnancy. Timing of intervention with senolytics (eg, dasatinib, quercetin, fisetin) is thus critical. Consequently, treatment windows for the use of senolytics, resulting in the elimination of senescent cells, need to be carefully defined. These include (1) the postpartum period, when senescent burden may still be modifiable; (2) the interpregnancy interval, to reduce risk of preeclampsia in subsequent pregnancies; and (3) postreproductive years, to mitigate long-term cardiovascular risk. Safety considerations are paramount, particularly in women of reproductive age. These agents trigger apoptosis selectively in senescent cells, while sparing nonsenescent cells, are effective immediately, and can be administered intermittently. A single dose (or a short duration of treatment) has been shown to be sufficient to cause the apoptosis of senescent cells and decrease senescent cell burden. Therefore, to avoid potential embryotoxic effects, senolytics can be administered after the affected pregnancy and before planning the next pregnancy, while still reaching the desired effect, that is, selective removal of senescent cells. In addition, given their short elimination half-lives, the risk of adverse effects for future pregnancies will be minimal. Senomorphics (eg, metformin, steroids, resveratrol), on the other hand, lead to suppression of SASP production by senescent cells without inducing apoptosis, and, as such, need to be administered throughout pregnancy to modify the secretory behavior of the senescent cells. One notable example in this class is metformin, which can be used during pregnancy given its proven safety record based on its use in diabetic pregnant patients. A systematic review and meta-analysis of randomized clinical trials has demonstrated that metformin use in women with either gestational diabetes or obesity is associated with reduced incidence of hypertensive disorders of pregnancy

compared with other treatments or placebo.¹⁶ These beneficial effects were ascribed to improved cardiovascular function, reduced gestational weight gain, and its known stabilization of vasoactive mediators; its senomorphic effects were ignored. Similarly, while it has been well recognized that offspring from preeclampsia pregnancies are at risk for future hypertension³⁴ and metabolic problems, the specific role of senescence and long-standing effects from in utero exposure to SASP is unknown.

Beyond pharmacological interventions, identification of senescence biomarkers may enable risk stratification and targeted prevention strategies. One notable example would be the use of SASP to guide the timing and dosing of senolytics before and senomorphic agents during pregnancies at risk for preeclampsia. Finally, as women with a history of preeclampsia develop CVD at higher rates and earlier in life³ compared with their normotensive pregnancy counterparts and demonstrate less favorable profiles of senescence markers,²⁹ integrating pregnancy history and senescence biology into cardiovascular risk assessment could transform care for women with a history of preeclampsia.



Knowledge Gaps and Future Directions

Despite compelling evidence, several key questions remain. Longitudinal studies are needed to determine the persistence of senescence markers after pregnancy and their relationship to clinical outcomes. The safety and efficacy of senolytic therapies in women must be rigorously evaluated, as ethical considerations surrounding interventions in reproductive-age women require careful attention.

Future research should also explore interactions between senescence and other aging pathways, including mitochondrial dysfunction, immune aging, and metabolic dysregulation. Understanding these interactions will be essential for developing comprehensive prevention strategies. In future research, we will use our established preeclampsia model, Interleukin-10 knock-out mice, to study the role of senolytics and senomorphics in preventing preeclampsia and in restoring normotensive pregnancy phenotypes.²⁵

CONCLUSIONS

Preeclampsia is increasingly recognized as a disorder with lifelong cardiovascular consequences. Accumulating evidence supports cellular senescence as a central mechanism linking pregnancy complications to accelerated biological aging and future CVD. Accelerated epigenetic aging, increased senescence burden across multiple tissues, MSC dysfunction, and reversibility with senolytic therapy collectively suggest that senescence is not merely a marker but also a mediator of disease. Targeting senescence pathways represents a promising

strategy to reduce cardiovascular risk and improve long-term health in women affected by preeclampsia.

ARTICLE INFORMATION

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Disclosures

None.

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